The deterministic part may however be approximated in various ways. We may solve it by splitting into two (volume preserving) parts, integrating first $\dot{\boldsymbol{q}} = \xi \boldsymbol{M}^{-1}\boldsymbol{p}$, where ξ and p are fixed (thus a drift along a given direction), followed by solving $\dot{\xi} = -(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U/\mu$, which is a similar drift in the auxiliary variable. Composing these drifts successively in a leapfrog approach, we arrive at the deterministic sequence:

$$\begin{aligned}
\xi_{n+1/2} &= \xi_n - (h/2)(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q}_n)/\mu, \\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\xi_{n+1/2}\boldsymbol{M}^{-1}\boldsymbol{p}, \\
\xi_{n+1} &= \xi_{n+1/2} - (h/2)(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q}_{n+1})/\mu.
\end{aligned}$$

Coupling this to the stochastic evolution computed from solving the OU parts in both ξ and $\boldsymbol{p}$ yields an alternative to Langevin dynamics that is 2nd order accurate in its steady state distribution and long term averages. Let us refer to this as the stochastic line sampling with leapfrog method.

Going a step further we may again obtain methods which exactly preserve the canonical measure based on canonical coordinate splitting, resulting in systems of the form

$$\dot{q}_i = \xi m_i^{-1} p_i, \qquad \dot{\xi} = -p_i U_{q_i}/(\mu m_i).$$

Up to time-rescaling by p_i, these are in fact the same types of systems as we encountered previously in our treatment of J-splitting of the Hamiltonian system. If desired, an rth order area-preserving approximation of the q_i, p_i dynamics could be substituted. This method results in paths that differ slightly from the linear path in the direction $\boldsymbol{M}^{-1}\boldsymbol{p}$. In simulations we found that this method, which we term "stochastic line sampling with coordinate splitting," was more stable than the leapfrog version when implemented using a further leapfrog splitting in each coordinate plane. However, we emphasize that both methods are highly experimental and are presented here as illustrations of the flexibility available in constructing methods to sample the canonical distribution—they have not yet been shown to be beneficial for molecular sampling applications.

We compare the trajectories and invariant distributions of the non-Newtonian line sampling thermostat (8.33)–(8.35) with three Newtonian-based thermostats: Langevin dynamics (7.4), Nosé-Hoover (8.2)–(8.4) and Nosé-Hoover-Langevin (8.27)–(8.29) dynamics. We seek to canonically sample a one-dimensional system with double-well potential energy function given as

$$U(q) = (q^2 - 1)^2.$$

Choosing all arbitrary system parameters to be one (i.e. setting $\beta = M = \gamma = \gamma' = \mu = 1$ for the methods), we select initial conditions $q(0) = p(0) = 1$ and $\xi(0) = 0.5$ with a timestep parameter $\delta t = 0.01$ sufficiently small to eliminate the possibility of numerical errors warping the results.